ESE5023 Assignment 02 Report

Ruochen, Liu 12631508

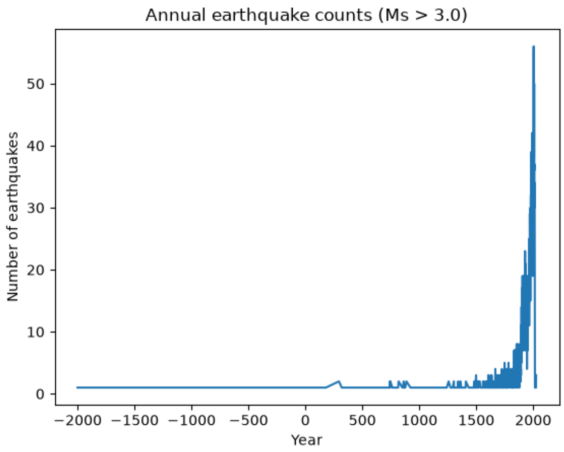
- PS2_1

1.1

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Country
CHINA          2139209.0
TURKEY         1199720.0
IRAN           1014453.0
ITALY           492236.0
SYRIA           419826.0
HAITI           323484.0
AZERBAIJAN      319251.0
JAPAN           242516.0
ARMENIA         177890.0
PAKISTAN        145085.0
IRAQ            136200.0
ECUADOR         135496.0
TURKMENISTAN    117412.0
PERU            102309.0
ISRAEL           93388.0
PORTUGAL         83572.0
GREECE           81162.0
CHILE            64277.0
INDIA            63507.0
TAIWAN           57154.0
Name: Deaths, dtype: float64
```

1.2

```
Year
-2000.0    1
-479.0     1
-426.0     1
-400.0     1
-373.0     1
          ..
 2015.0    1
 2017.0    1
 2019.0    1
 2024.0    1
 2025.0    3
Length: 511, dtype: int64
```



关于 trend：图中明显可以看出近代记录的地震次数明显增加，但这不一定代表实际地震变多，考虑到早期记录不完整，现代监测和报告系统更完善，因此上升趋势可能一方面是地震频次确实增加了另一方面也反映数据记录变得更全面。

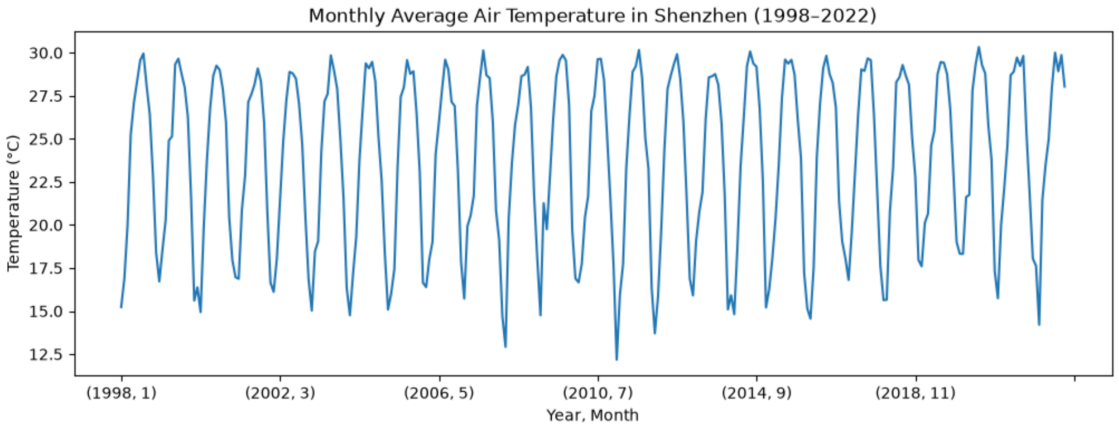
1.3

	Country	Count	Year	Mo	Dy	Location Name
0	CHINA	626	1920	12	16	CHINA: GANSU PROVINCE, SHANXI PROVINCE
1	JAPAN	429	869	7	13	JAPAN: SANRIKU
2	JAPAN	429	887	8	26	JAPAN: NANKAIDO
3	INDONESIA	427	2004	12	26	INDONESIA: SUMATRA: ACEH: OFF WEST COAST
4	IRAN	385	856	12	22	IRAN: DAMGHAN, QUMIS
5	TURKEY	359	1939	12	26	TURKEY: ERZINCAN
6	TURKEY	359	1999	8	17	TURKEY: ISTANBUL, KOCAELI, SAKARYA
7	ITALY	338	1915	1	13	ITALY: MARSICA, AVEZZANO, ABRUZZI
8	GREECE	296	365	7	21	GREECE: CRETE: KNOSSOS
9	GREECE	296	1303	8	8	GREECE: CRETE
10	USA	283	1957	3	9	ALASKA
11	PHILIPPINES	233	1897	9	21	PHILIPPINES: MINDANAO, ZAMBOANGA, SULU, ISABELA
12	MEXICO	217	1787	3	28	MEXICO: SAN MARCOS, OAXACA
13	MEXICO	217	1907	4	15	MEXICO: GUERRERO
14	CHILE	200	1730	7	8	CHILE: VALPARAISO
15	PERU	195	1619	2	14	PERU: TRUJILLO, PIURA, SANTA
16	PERU	195	1942	8	24	PERU: ICA, NAZCA
17	RUSSIA	159	1952	11	4	RUSSIA: KAMCHATKA PENINSULA
18	PAPUA NEW GUINEA	107	2000	11	16	PAPUA NEW GUINEA: NEW IRELAND, DUKE OF YORK
19	TAIWAN	103	1920	6	5	TAIWAN
20	TAIWAN	103	1966	3	12	TAIWAN
21	INDIA	102	1897	6	12	INDIA: ASSAM; BANGLADESH
22	COLOMBIA	83	1904	1	20	PANAMA-COLOMBIA
23	AFGHANISTAN	72	1909	7	7	AFGHANISTAN: HINDU-KUSH
24	NEW ZEALAND	72	1826	<NA>	<NA>	NEW ZEALAND: FJORDLAND
25	NEW ZEALAND	72	1855	1	23	NEW ZEALAND: WELLINGTON,WAIOURU,WANGANUI,OTAKI
26	ECUADOR	70	1906	1	31	ECUADOR: OFF COAST
27	VENEZUELA	65	1900	10	29	VENEZUELA: MACUTO
28	VANUATU	65	1910	6	16	VANUATU ISLANDS
29	VANUATU	65	1913	10	14	VANUATU ISLANDS
30	SOLOMON ISLANDS	63	1934	7	18	SOLOMON ISLANDS: SANTA CRUZ ISLANDS
31	SOLOMON ISLANDS	63	1977	4	21	SOLOMON ISLANDS
32	ALGERIA	58	1980	10	10	ALGERIA: NORTHERN
33	ALBANIA	56	1893	6	14	ALBANIA: HIMARA, DHERMI, KUC, KUDHESI, VLORE, ...
34	PAKISTAN	55	1945	11	27	PAKISTAN: MAKRAN COAST
35	CROATIA	53	1667	4	6	BALKANS NW: CROATIA: DUBROVNIK: RAGUSA
36	GUATEMALA	47	1942	8	6	GUATEMALA: NEAR S COAST
37	FRANCE	43	1909	6	11	FRANCE: VERNEGUES,CHARLEVAL,LA ROQUE D'ANTHERON
38	MYANMAR (BURMA)	43	1912	5	23	MYANMAR (BURMA): MANDALAY, MOGOK, MAYMYO
39	MYANMAR (BURMA)	43	2025	3	28	MYANMAR (BURMA); THAILAND
40	EL SALVADOR	42	1915	9	7	EL SALVADOR
41	EL SALVADOR	42	2001	1	13	EL SALVADOR; GUATEMALA
42	NICARAGUA	40	1898	4	29	NICARAGUA: LEON, CHINANDEGA, MANAGUA

43	ARGENTINA	39	1894	10	27	ARGENTINA: LA RIOJA, SAN JUAN, MENDOZA
44	ARGENTINA	39	1912	12	7	ARGENTINA: SANTIAGO DEL ESTERO PROVINCE
45	USA TERRITORY	39	1902	9	22	GUAM: AGANA
46	COSTA RICA	37	1950	10	5	COSTA RICA-NICARAGUA
47	COSTA RICA	37	2012	9	5	COSTA RICA: NICOYA
48	SPAIN	35	1884	12	25	SPAIN: ARENAS DEL REY, VEGA, ALHAMA, MALAGA
49	SYRIA	34	1202	5	20	SYRIA: SOUTHWESTERN
50	AUSTRALIA	34	1989	5	23	MACQUARIE ISLAND: MACQUARIE STATION; NEW ZEALAND
51	PORTUGAL	29	1761	3	30	PORTUGAL: LISBON, PORTO
52	TAJIKISTAN	29	1907	10	21	TAJIKISTAN: KARATAG
53	TAJIKISTAN	29	1911	2	18	TAJIKISTAN: USOY, BARCHIDIV, POSOR, NISUR
54	TAJIKISTAN	29	1949	7	10	TAJIKISTAN
55	AZORES (PORTUGAL)	28	1980	1	1	AZORES: TERCEIRA, ANGRA DO HEROISMO
56	NEW CALEDONIA	28	1875	3	28	NEW CALEDONIA: LOYALTY ISLANDS: LIFOU ISLAND
57	TONGA	26	1919	4	30	TONGA ISLANDS
58	ISRAEL	25	1759	10	30	ISRAEL: ZEFAT (SAFED)
59	IRAQ	24	1864	12	2	IRAQ: ZURBATIYAH, BADRAH, TURSAQ, BAGHDAD
60	PANAMA	24	1882	9	7	PANAMA: SAN BLAS ARCHIPELAGO
61	KERMADEC ISLANDS (NEW ZEALAND)	23	1986	10	20	KERMADEC ISLANDS: RAOUL
62	SOUTH KOREA	22	1643	7	25	SEA OF JAPAN
63	SOUTH KOREA	22	1649	12	9	YELLOW SEA
64	SOUTH KOREA	22	1700	9	12	SOUTH KOREA: CHUNGCHONG
65	SLOVENIA	22	1511	3	26	BALKANS NW: SLOVENIA: IDRIJA,SKOFJA LOKA
66	CANADA	22	1949	8	22	CANADA: QUEEN CHARLOTTE ISLANDS
67	NEPAL	21	1505	6	6	NEPAL-INDIA
68	MOROCCO	21	2004	2	24	MOROCCO: AL HOCEIMA, IMZOURENE, BENI ABDALLAH
69	HAITI	21	1842	5	7	HAITI: CAP-HAITIEN
70	FIJI	20	1919	1	1	FIJI ISLANDS
71	DOMINICAN REPUBLIC	19	1946	8	4	DOMINICAN REPUBLIC: NORTHEASTERN COAST
72	BOLIVIA	19	1916	8	25	BOLIVIA-NORTHERN CHILE
73	BOLIVIA	19	1959	6	14	BOLIVIA-NORTHERN CHILE
74	JAMAICA	19	1899	6	14	JAMAICA
75	BANGLADESH	19	1918	7	8	BANGLADESH: SRIMANGAL
76	BULGARIA	18	1904	4	4	BULGARIA: STRUMA
77	EGYPT	17	1995	11	22	EGYPT: NUWAYBI; SAUDI ARABIA; ISRAEL; JORDAN
78	AZERBAIJAN	17	1667	11	<NA>	AZERBAIJAN: SHEMAKHA (SAMAXI)
79	AZERBAIJAN	17	1902	2	13	AZERBAIJAN: SEMACHA; N IRAN
80	ICELAND	17	2000	6	17	ICELAND: VESTMANNAEYJAR, HELLA
81	ICELAND	17	2000	6	21	ICELAND: GRIMSNES, SELFOSS, EYRARBAKKI, STOKK...
82	KYRGYZSTAN	15	1911	1	3	KAZAKHSTAN: ALMA-ATA, TURKESTAN; AFGHANISTAN
83	GEORGIA	15	1905	10	21	GEORGIA: CAUCASUS
84	UK	15	1931	6	7	UK: SCARBOROUGH, GRIMSBY
85	SERBIA	15	1980	5	18	BALKANS NW: SERBIA
86	ROMANIA	15	1986	8	30	ROMANIA: BUCHAREST, KISHINEV-KAGUL
87	HONDURAS	15	1856	8	4	HONDURAS: COAST
88	HONDURAS	15	1910	1	1	HONDURAS
89	CUBA	15	1992	5	25	CUBA: PILON, MANZANILLO
90	LEBANON	14	1759	11	25	LEBANON-SYRIA: BAALBEC; SYRIA: DAMASCUS, ANTIOCH
91	ARMENIA	14	1988	12	7	ARMENIA: LENINAKAN, SPITAK, KIROVAKAN
92	UZBEKISTAN	14	1976	4	8	UZBEKISTAN: GAZLI
93	UZBEKISTAN	14	1976	5	17	UZBEKISTAN: GAZLI, BUKHARA
94	UZBEKISTAN	14	1984	3	19	UZBEKISTAN: GAZLI
95	SOUTH AFRICA	14	1942	11	10	SOUTH AFRICA: PRINCE EDWARD ISLAND
96	NORTH MACEDONIA	13	1979	5	24	BALKANS NW: MACEDONIA: DEBAR
97	BRAZIL	13	1986	11	30	BRAZIL: JOAO CAMARA, NATAL
98	TURKMENISTAN	12	1895	7	8	TURKMENISTAN: UZUN-ADA
99	UKRAINE	12	1927	9	11	UKRAINE: CRIMEA: SEBASTOPOL
100	BOSNIA-HERZEGOVINA	11	1990	11	27	BALKANS NW: BOSNIA-HERZEGOVINA: TITOGRAD
101	ETHIOPIA	11	1961	6	1	ETHIOPIA: KARAKORE
102	CYPRUS	10	1953	9	10	CYPRUS: PAPHOS
103	YEMEN	10	1982	12	13	YEMEN: DHAMAR
104	MONTENEGRO	10	1979	4	15	BALKANS NW: MONTENEGRO
105	MARTINIQUE	10	1906	12	3	MARTINIQUE
106	TUNISIA	9	1957	2	20	TUNISIA: SIDI ABID,SIDI TOUIL (LA MEDJA),CAIL...
107	KAZAKHSTAN	9	1978	3	24	KAZAKHSTAN: ALMA-ATA
108	GERMANY	9	1978	9	3	GERMANY
109	GUADELOUPE	9	1843	2	8	GUADELOUPE: POINTE-A-PITRE
110	AUSTRIA	8	1998	4	12	AUSTRIA: ARNOLDSTEIN; SLOVENIA: BOVEC, KOBARID
111	TRINIDAD AND TOBAGO	8	1831	12	3	TRINIDAD & ST. CHRISTOPHER
112	TANZANIA	8	1910	12	13	TANZANIA: RUKWA
113	SAMOA	8	1917	6	26	SAMOA ISLANDS
114	POLAND	8	2019	7	1	POLAND: KATOWICE
115	ATLANTIC OCEAN	7	1941	11	25	ATLANTIC OCEAN: NORTHERN
116	MONGOLIA	7	1905	7	23	MONGOLIA
117	CONGO	7	2005	12	5	CONGO: KALEMIE
118	SOUTH GEORGIA AND THE SOUTH SANDWICH ISLANDS	7	1929	6	27	SOUTH SANDWICH ISLANDS
119	ANTARCTICA	7	1998	3	25	BALLENY ISLANDS
120	NORTH KOREA	6	1668	7	31	NORTH KOREA: YELLOW SEA
121	GHANA	6	1939	6	22	GHANA: ACCRA
122	ERITREA	6	1915	9	23	ERITREA: ASMARA
123	VIETNAM	6	1983	6	24	VIETNAM
124	JORDAN	5	362	5	24	JORDAN: AL-KARAK
125	MICRONESIA, FED. STATES OF	5	1911	8	16	MICRONESIA, FED. STATES OF: CAROLINE ISLANDS
126	RWANDA	5	2008	2	14	RWANDA: GISENYI
127	BHUTAN	5	2009	9	21	BHUTAN: TASHIGANG
128	THAILAND	4	1983	4	22	THAILAND: BANGKOK
129	SOUTH SUDAN	4	1990	5	20	SOUTH SUDAN: JUBA, MAYA; UGANDA: NAKURA
130	UGANDA	4	1966	3	20	UGANDA: KICHWAMBA, BONDIBOGYO; TANZANIA; DR ...
131	MALAWI	4	1989	3	10	MALAWI: SALIMA, DEDZA, MOHINJI
132	NETHERLANDS	4	1992	4	13	THE NETHERLANDS: ROERMOND; GERMANY: BONN, HEIN...
133	ANTIGUA AND BARBUDA	3	1690	4	16	ANTIGUA; SAINT KITTS AND NEVIS
134	ANTIGUA AND BARBUDA	3	1974	10	8	ANTIGUA AND BARBUDA; ST KITTS
135	KENYA	3	1928	1	6	KENYA: SUBUKIA
136	INDIAN OCEAN	3	1928	3	9	INDIAN OCEAN: S
137	MALAYSIA	3	1976	7	26	MALAYSIA: SABAH: LAHAD,DATU,KANAK
138	SAUDI ARABIA	3	2009	5	19	SAUDI ARABIA: WESTERN
139	MOZAMBIQUE	3	2006	2	22	MOZAMBIQUE
140	LIBYA	2	1963	2	21	LIBYA: BARCE (AL MARJ)
141	NORWAY	2	1819	8	31	NORWAY: RANA REGION: LUROY
142	UK TERRITORY	2	1983	11	30	INDIAN OCEAN: CHAGOS ARCHIPELAGO: DIEGO GARCIA
143	SOLOMON SEA	2	1895	3	6	W. SOLOMON SEA
144	DJIBOUTI	2	1989	8	20	DJIBOUTI: GALAFI, YABAKI; ETHIOPIA
145	PACIFIC OCEAN	2	1958	11	4	EAST PACIFIC RIDGE
146	LAOS	2	2007	5	16	LAOS: BOKEO; THAILAND: CHIANG RAI, CHIAN SAEN
147	FRENCH POLYNESIA	1	1848	7	12	FRENCH POLYNESIA: TAHITI
148	TASMAN SEA	1	1892	1	26	TASMAN SEA
149	PALAU	1	1914	10	23	MICRONESIA, FED. STATES OF: CAROLINE ISLANDS
150	GABON	1	1974	9	23	GABON
151	GUINEA	1	1983	12	22	GUINEA: GAOUAL-KOUMBIA
152	BERING SEA	1	1991	2	21	BERING SEA
153	WALLIS AND FUTUNA (FRENCH TERRITORY)	1	1993	3	12	FUTUNA ISLAND
154	SUDAN	1	1993	8	1	SUDAN: KHARTOUM

- **PS2_2**

在 user guide 中，POS88-92 表示，数据展示的 scaling factor 是 10，也就是在计算的时候要除以 10 才能得到以摄氏度为单位的温度值，并且+9999 表示该值缺失，所以筛选的时候也要排除这些，POS93 表示的是数据中的温度质量码，可以看到不同的数字和字母代表不同的含义，我在这里只筛选了温度质量码为 1 或者 5 的数值，选择的原因基于数据都为 pass the quality control check，意思是这些数据都通过的所有的质量检查。其他的字母有的表示数据可疑或者错误在筛选时不需要，因为如果数据本身存在问题的话那做图出来其实不具有参考意义。



根据上面的图可以看出这里其实是存在一些 trend，整体看的话波动上下幅度都比较类似，每年都会上下浮动，平均最低温可能在 15-18 度附近，少数年份可以到 12-15 度左右平均最高温在 30 度左右。

- **PS2_3**

这里先选前 17 列，因为本题后面要用的 SID、NAME、BASIN、ISO_TIME、LAT、LON 和 WMO_WIND 都在其中。skiprows=[1, 2] 跳过单位说明行和额外说明行；parse_dates 将时间列读成日期格式；na_values 把指定的占位文字识别为缺失值。

3.1

Collapse Output

	SID	SEASON	NUMBER	BASIN	SUBBASIN	NAME	ISO_TIME	NATURE	LAT	LON	WMO_WIND	WMO_PRES	WMO_AGENCY	TRACK_TYPE	DIST2LAND	LANDFALL	IFLAG
0	1842298N11080	1842	1	NI	BB	NaN	1842-10-25 06:00:00	NR	10.8709	79.8265				main	0	0	_____P___
1	1842298N11080	1842	1	NI	BB	NaN	1842-10-25 09:00:00	NR	10.8431	79.3524				main	0	0	_____P___
2	1842298N11080	1842	1	NI	BB	NaN	1842-10-25 12:00:00	NR	10.8188	78.8772				main	0	0	_____P___
3	1842298N11080	1842	1	NI	BB	NaN	1842-10-25 15:00:00	NR	10.8000	78.4000				main	0	0	_____O___
4	1842298N11080	1842	1	NI	AS	NaN	1842-10-25 18:00:00	NR	10.7884	77.9194				main	0	0	_____P___

	NAME	WMO_WIND
12921	PATRICIA	185.0
9087	ALLEN	165.0
10011	GILBERT	160.0
11944	WILMA	160.0
11067	LINDA	160.0
13307	DORIAN	160.0
12337	RICK	155.0
13098	IRMA	155.0
11927	RITA	155.0
11190	MITCH	155.0

3.2

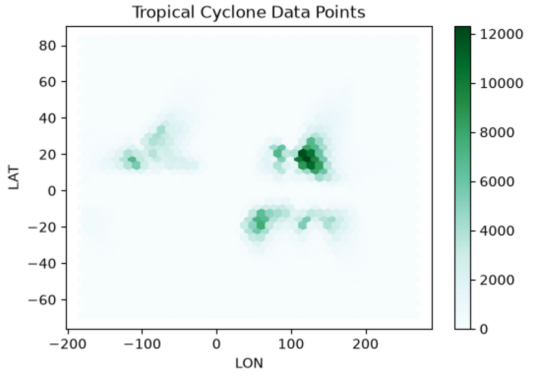
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12921    185.0
9087     165.0
10011    160.0
11944    160.0
11067    160.0
13307    160.0
12337    155.0
13098    155.0
11927    155.0
11190    155.0
8722     150.0
12945    150.0
7532     150.0
12107    150.0
11908    150.0
10476    150.0
6311     150.0
10704    150.0
8985     150.0
5881     150.0
Name: WMO_WIND, dtype: float64
```

3.3

```
BASIN
WP    236576
SI    160668
SP     67119
EP     62412
NI     55401
SA       119
Name: count, dtype: int64
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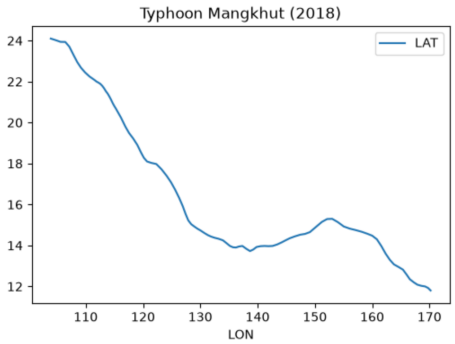
3.4

```
<Axes: title={'center': 'Tropical Cyclone Data Points'}, xlabel='LON', ylabel='LAT'>
```



3.5

```
<Axes: title={'center': 'Typhoon Mangkhut (2018)'}, xlabel='LON'>
```

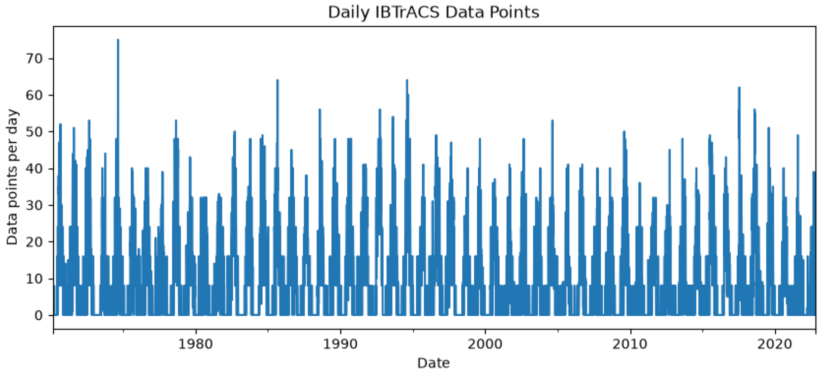


3.6

```
<class 'pandas.DataFrame'>
Index: 176352 entries, 350393 to 707175
Data columns (total 17 columns):
 #   Column      Non-Null Count   Dtype
---  ------      --------------   -----
 0   SID         176352 non-null  str
 1   SEASON      176352 non-null  int64
 2   NUMBER      176352 non-null  int64
 3   BASIN       176352 non-null  str
 4   SUBBASIN    176352 non-null  str
 5   NAME        157891 non-null  str
 6   ISO_TIME    176352 non-null  datetime64[us]
 7   NATURE      176352 non-null  str
 8   LAT         176352 non-null  float64
 9   LON         176352 non-null  float64
 10  WMO_WIND    52023 non-null   float64
 11  WMO_PRES    176352 non-null  str
 12  WMO_AGENCY  176352 non-null  str
 13  TRACK_TYPE  176352 non-null  str
 14  DIST2LAND   176352 non-null  int64
 15  LANDFALL    176352 non-null  str
 16  IFLAG       176352 non-null  str
dtypes: datetime64[us](1), float64(3), int64(3), str(10)
memory usage: 32.9 MB
```

3.7

```
<Axes: title={'center': 'Daily IBTrACS Data Points'}, xlabel='Date', ylabel='Data points per day'>
```



3.8

```
ISO_TIME
1      1.596154
2      1.384615
3      1.423077
4      1.788462
5      2.019231
         ...
362    3.038462
363    2.538462
364    2.000000
365    1.788462
366    1.000000
Length: 366, dtype: float64
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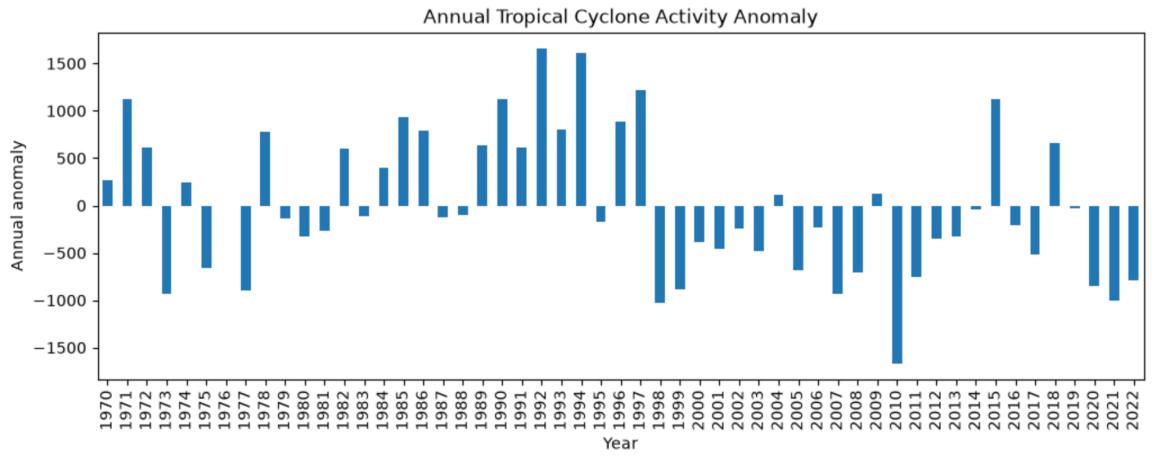
3.9

```
ISO_TIME
1970-02-19     7.018868
1970-02-20     7.094340
1970-02-21     7.188679
1970-02-22     7.377358
1970-02-23     7.490566
                 ...
2022-10-08   -17.245283
2022-10-09   -17.301887
2022-10-10   -12.396226
2022-10-11   -18.528302
2022-10-12   -14.924528
Freq: D, Length: 19229, dtype: float64
```

3.10

从图上看，1992–1994 年的正异常最突出，其中 1992 年最高，1998 年和 2015 年也明显偏高，负异常较高的年份看得出是 2010 年，以及 2020–2022 年，表示这些年份的数据点数低于按日序计算的平均水平。

```
<Axes: title={'center': 'Annual Tropical Cyclone Activity Anomaly'}, xlabel='Year', ylabel='Annual anomaly'>
```



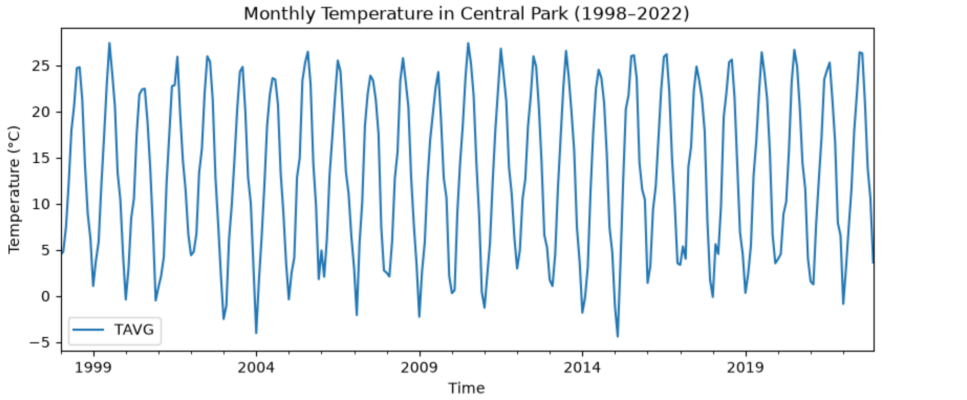
- **PS2-4**

4.1

```
DATE    0
TAVG    0
dtype: int64
<class 'pandas.DataFrame'>
RangeIndex: 300 entries, 0 to 299
Data columns (total 3 columns):
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
 0   DATE             300 non-null    datetime64[us]
 1   TAVG             300 non-null    float64
 2   TAVG_ATTRIBUTES  300 non-null    str
dtypes: datetime64[us](1), float64(1), str(1)
memory usage: 7.7 KB
None
```

4.2

```
<Axes: title={'center': 'Monthly Temperature in Central Park (1998–2022)'}, xlabel='Time', ylabel='Temperature (°C)'>
```



4.3

```
Mean: 13.268066666666666
Median: 13.094999999999999
Minimum: -4.41
Maximum: 27.42
Standard deviation: 8.624258831741919
```

以上分别计算的是平均值，中位数，最小值，最大值，以及标准差